

**SIMATS ENGINEERING  
ASSESSMENT TOOL 2**

**COURSE CODE: CSA1402**

**COURSE NAME: Compiler Design**

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**COURSE OUTCOME COVERED**

*CO2: Construct top-down and bottom up parsers using CFG for parsing conflicts and error recovery for efficient syntax tree generation (BL3)*

*Assessment Tool Weightage: 20%*

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**QUESTIONS: 20**

**Total Marks:100**

Annexure A

**SHORT ANSWER TEST**

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**Code of Conduct:**

I Akshaya.A/192421222 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.  
I understand that any violation of academic integrity rules will result in disciplinary action.

**Signature of the student:** Akshaya.A

| Criteria                  | Weightage | Level 4 – Exemplary | Level 3 – Proficient | Level 2 – Developing | Level 1 – Beginning |
|---------------------------|-----------|---------------------|----------------------|----------------------|---------------------|
| Understanding of Concepts | 25%       |                     |                      |                      |                     |
| Explanation               | 25%       |                     |                      |                      |                     |
| Application               | 20%       |                     |                      |                      |                     |
| Organization              | 15%       |                     |                      |                      |                     |
| Accuracy                  | 10%       |                     |                      |                      |                     |
| Clarity                   | 5%        |                     |                      |                      |                     |

Faculty Incharge

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DATE: 04.08.2026

COURSE: CSA1402 COMPILER DESIGN

CO2-AT2 - SHORT ANSWER TEST

1. Construct language for  $(a|b)^* a(a|b)|(a|b)$

2. a) Predictive Parser

$$S \rightarrow (L) | a$$

$$L \rightarrow L, S / S$$

81  
100

Ans:

Left Recursion:  $S \rightarrow (L) | a$

$$L \rightarrow SL'$$

$$L' \rightarrow , SL' / \epsilon$$

$$\text{FIRST}(S) \rightarrow \{ (, a \}$$

$$\text{FIRST}(L) \rightarrow \{ (, a \}$$

$$\text{FIRST}(L') \rightarrow \{ , , \epsilon \}$$

$$\text{FOLLOW}(S) \rightarrow \{ \$ \}$$

$$\text{FOLLOW}(L) \rightarrow \{ \$, ) \}$$

$$\text{FOLLOW}(L') \rightarrow \{ \$, ) \}$$

Non Terminals

id

(

)

,

$\epsilon$   $\$$

S

$$S \rightarrow (L)/a$$

L

$$L \rightarrow SL'$$

L'

$$L' \rightarrow ,SL'$$

$$L' \rightarrow \epsilon$$

3) Predictive Parser  $S \rightarrow SS+ / SS^* / a$

Input string: "aa+a"

Ans:

Left Recursion:

$$S \rightarrow as'$$

$$s' \rightarrow s+s' / s*s' / \epsilon$$

$$\text{FIRST}(S) \rightarrow \{a\}$$

$$\text{FIRST}(s') \rightarrow \{a, \epsilon\}$$

$$\text{FOLLOW}(S) \rightarrow \{ \$, +, * \}$$

$$\text{FOLLOW}(s') \rightarrow \{ \$, +, * \}$$

Non Terminals

id

+

Terminals

\*

$\$$

$\$$

S

$$S \rightarrow as'$$

$$S \rightarrow as'$$

$$\$ \rightarrow as'$$

$$S \rightarrow as'$$

s'

$$s' \rightarrow s+s'$$

$$s' \rightarrow s+s'$$

$$s' \rightarrow \epsilon$$

Stack

Input

output

|                 |                   |                         |
|-----------------|-------------------|-------------------------|
| \$ S            | aa + a            | $S \rightarrow as'$     |
| \$ as'          | aa + a            | $s' \rightarrow s + s'$ |
| \$ as + s'      | aa + a            | $S \rightarrow as'$     |
| \$ + s' aas'    | aa + a            |                         |
| \$ + s' s' aa   | <del>aa</del> + a |                         |
| \$ s' + s'      | + a               | $s' \rightarrow s + s'$ |
| \$ s' s' s + s' | * a               | $S \rightarrow as'$     |
| \$ s' as' + s'  | a                 |                         |
| \$ s' + s' s'   | Accept            |                         |

4) Left Recursion  $S \rightarrow Sa | Sb | c$

Ans:

$$S \rightarrow cs'$$

$$s' \rightarrow as' / bs' / \epsilon$$

## 5) Predictive Parsing

$$A \rightarrow abcD / abcE / f$$

Left Recursion:

$$A \rightarrow fA'$$

$$A' \rightarrow bcD / A'$$

$$A' \rightarrow Ecb / A'$$

$$A \rightarrow abcD / abcE / f$$

$$\text{FIRST}(A) \rightarrow \{ a, f \}$$

$$\text{FOLLOW}(A) \rightarrow \{ \$ \}$$

Non Terminal

A

Terminal

id

E

\$

$$A \rightarrow abcD / abcE / f$$

6. Degree - F = Degree - C \* 1.8 + 32

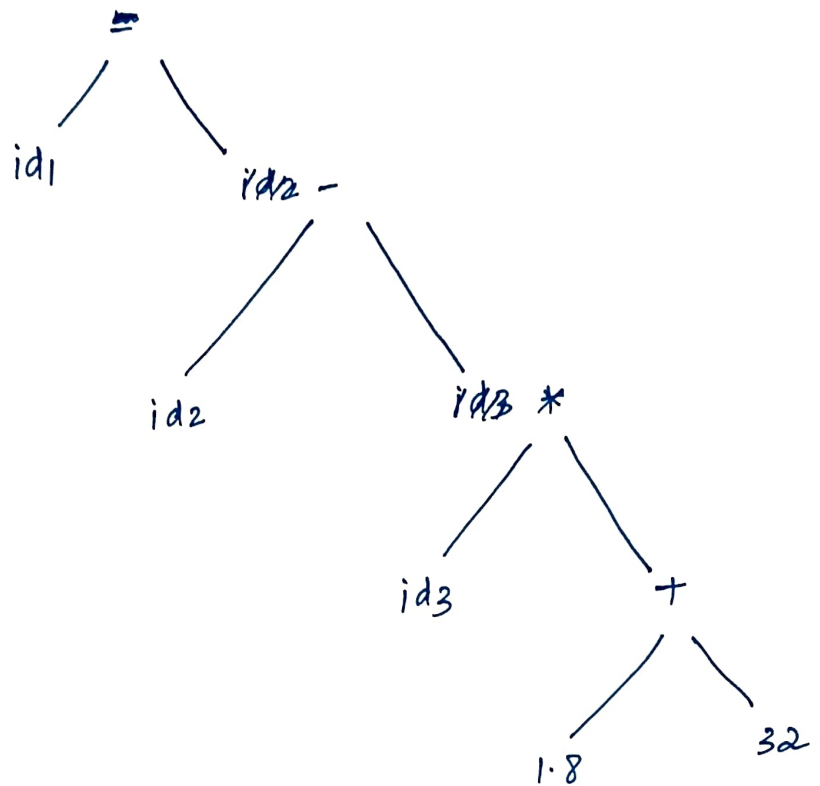
Ans:

Lexical Analyzer:

$$\langle id_1 \rangle (-) \langle id_2 \rangle = \langle id_1 \rangle (-) \langle id_3 \rangle (*) \langle 1.8 \rangle$$

$$\langle id_1 \rangle = \langle id_2 \rangle (-) \langle id_3 \rangle (*) \langle 1.8 \rangle (+) \langle 32 \rangle$$

# Syntax Analyzer :



Code Three Address Code :

temp1 = ~~temp~~ id2 - id3

temp2 = 1.8 + 32

temp3 = temp1 \* temp2

id1 = temp3

Code optimizer

temp3 = temp1 \* temp2

id1 = temp3

code generator:

movF → id3, r2

~~movF~~  
subF → id2, r2

movF → 32, r1

ADDF → 1.8, r1

~~movF~~ → r2, r1

5)

Left Factoring

$A \rightarrow abcd / abce / f$

$A \rightarrow abca'$

$A' \rightarrow D / E / f$

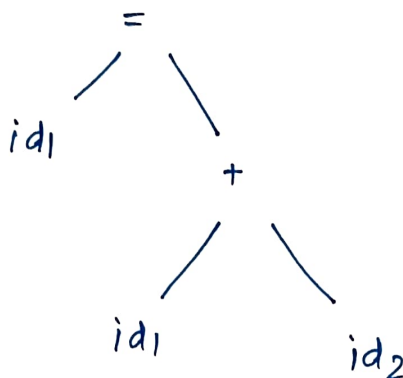
7)

# Phases of Compiler

sum = sum + i

Lexical Analyzer:  $\langle id_1 \rangle \langle = \rangle \langle id_1 \rangle (+) \langle id_2 \rangle$

Syntax Analyzer:



~~Semantic~~ Three Address Code:

temp1 = id1 + id2

temp2 = temp1

id1 = temp2

Code Optimizer: temp1 = id1 + id2

id1 = temp1

Code Generator:

mov F = id2, r2

ADD F = id1, r2

mov F = r2, id1

$$S \rightarrow aABb, A \rightarrow c/\epsilon \quad B \rightarrow d/\epsilon$$

$$S \rightarrow aABb$$

$$A \rightarrow c/\epsilon$$

$$B \rightarrow d/\epsilon$$

Left Recursion:

$$S \rightarrow aABb$$

$$A \rightarrow c/\epsilon$$

$$B \rightarrow d/\epsilon$$

$$\text{FIRST}(S) \rightarrow \{a\}$$

$$\text{FIRST}(A) \rightarrow \{c, \epsilon\}$$

$$\text{FIRST}(B) \rightarrow \{d, \epsilon\}$$

$$\text{FOLLOW}(S) \rightarrow \{\$ \}$$

$$\text{FOLLOW}(A) \rightarrow \{\$ \}$$

$$\text{FOLLOW}(B) \rightarrow \{\$ \}$$

|              |    | Terminal             |                   |                   |  |
|--------------|----|----------------------|-------------------|-------------------|--|
| Non-Terminal | id | a                    | c                 | d                 |  |
| S            |    | $S \rightarrow aABb$ |                   |                   |  |
| A            |    |                      | $A \rightarrow c$ |                   |  |
| B            |    |                      |                   | $B \rightarrow d$ |  |

Stack Implementation:

| stack  | Input | output               |
|--------|-------|----------------------|
| \$S    | acdb  | $S \rightarrow aABb$ |
| \$aABb | acdb  | $A \rightarrow c$    |
|        | acdb  | $B \rightarrow d$    |

\$adadb

acdb

\$

Accept

9) Lex program to recognize & count identifiers.

10) Rules for Left Recursion & Left Factoring.

Left Factoring : \* Given Production has to be evaluated  
\* Common strings has to be written first  
\* Uncommon & leftout strings to be written next.

Eg:  $A \rightarrow abcd / abce / f$

$A \rightarrow abca'$

$A' \rightarrow d / e / f$

Left Recursion: \* For the given production, take from last string and the new string has to be included [Eg:  $S \rightarrow ss+ / ss* / a$   
 $S \rightarrow as'$

\* From beginning string, write the strings for  $s'$   
 $s' \rightarrow s+s' / s*s' / \epsilon$

## Shift reduce Parser.

- \* Input string is given.
- \* \$ and the 1st production Non terminal is written.
- \* check whether the input string is greater than production string. If so, shift action done.
- \* Else Reduce the string.

Eg:  $E' \rightarrow id$

\$ E                      id + id \* id                      shift

\$ E id                      + id \* id                      Reduce  $E' \rightarrow id$

Here, id is greater than E,  $\therefore$  id is shifted

Now Reduce  $E' \rightarrow id$ , Because + is lesser than id

~~3~~ \$ E E'                      + id \* id

12)

$S \rightarrow pB$

$B \rightarrow qB/\eta$

Ans.

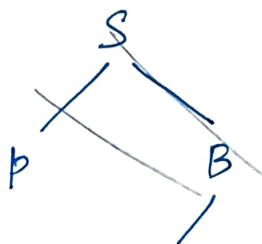
Leftmost Derivation:

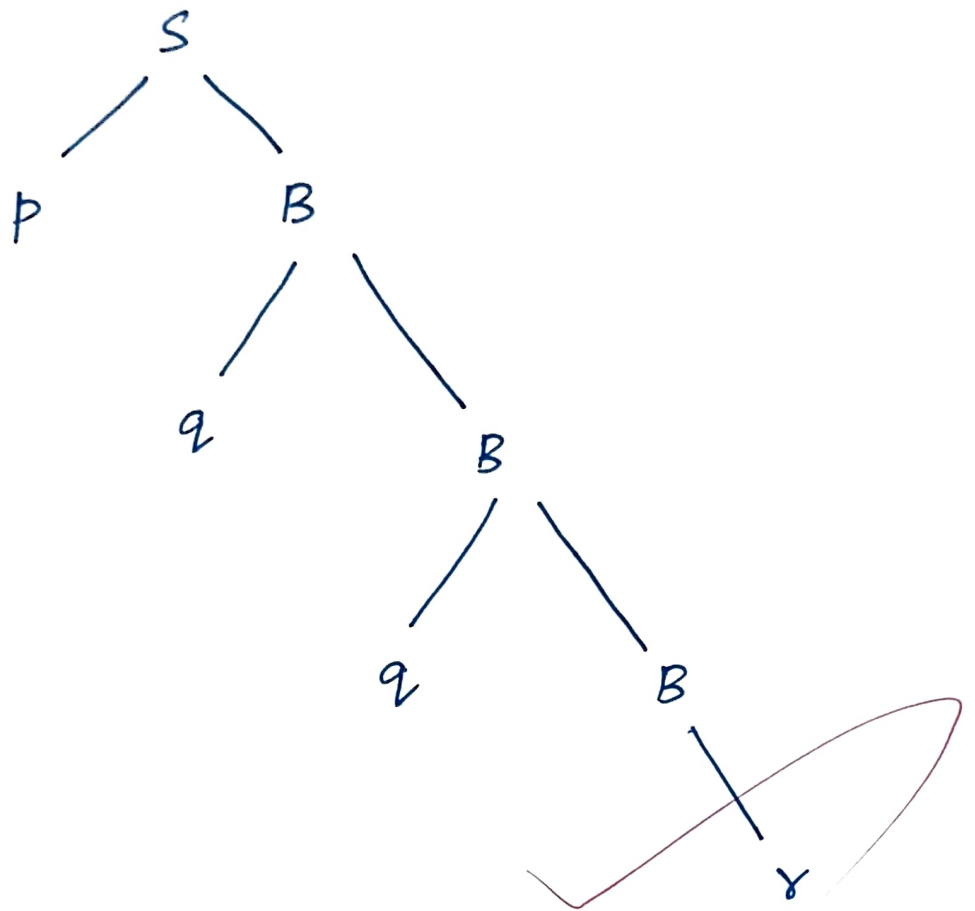
$S \rightarrow pB$

~~3~~  $\rightarrow pqB/\eta$

~~2~~  $\rightarrow pqqB/\eta$

$\rightarrow pqq\eta$





Therefore input string  $pqq r$  is derived .

13)

$$E \rightarrow E + T / T$$

$$T \rightarrow T * F / F$$

$$F \rightarrow (E) / id$$

Ans:

Left Recursion :

$$E \rightarrow TE'$$

$$E' \rightarrow +TE' / \epsilon$$

$$T \rightarrow FT'$$

$$T' \rightarrow *FT' / \epsilon$$

$$F \rightarrow (E) / id$$

FIRST (E)  $\rightarrow \{ (, id \}$

FIRST (E')  $\rightarrow \{ +, \epsilon \}$

FIRST (T)  $\rightarrow \{ (, id \}$

FIRST (T')  $\rightarrow \{ *, \epsilon \}$

FIRST (F)  $\rightarrow \{ (, id \}$

FOLLOW (E)  $\rightarrow \{ \$, ) \}$

FOLLOW (E')  $\rightarrow \{ \$, ) \}$

FOLLOW (T)  $\rightarrow \{ +, \$, ) \}$

FOLLOW (T')  $\rightarrow \{ +, \$, ) \}$

FOLLOW (F)  $\rightarrow \{ *, +, \$, ) \}$

4

14

| stack     |                      | Input  | shift / Reduce            |
|-----------|----------------------|--------|---------------------------|
| \$S       | $S \rightarrow AB$   | abcc\$ | shift                     |
| \$Sa      | $A \rightarrow aA a$ | bcc\$  | Reduce $S \rightarrow AB$ |
| \$ABa     | $B \rightarrow cB c$ | bcc\$  | Reduce $B \rightarrow cB$ |
| \$AacB    |                      | bcc\$  | shift                     |
| \$AacBb   |                      | cc\$   | Reduce $B \rightarrow cB$ |
| \$AacbcB  |                      | cc\$   | shift                     |
| \$AacbcBc |                      | c\$    | Reduce $B \rightarrow cB$ |

|            |     |                            |
|------------|-----|----------------------------|
| \$AacbccB  | c\$ | shift                      |
| \$AacbccBc | \$  | Reduce $Aab \rightarrow S$ |
| \$ScBccc   | \$  | Reduce $cB \rightarrow B$  |
| \$ScBccc   | \$  | Accept                     |

15) shift Reduce parsing  
 abbc  
 $S \rightarrow aAB$   
 $A \rightarrow bA/b$   
 $B \rightarrow c$

| Stack     | Input     | Output                     |
|-----------|-----------|----------------------------|
| \$S       | a*b*b*c\$ | shift                      |
| \$Sa      | *b*b*c\$  | Reduce $S \rightarrow aAB$ |
| \$aaAB    | *b*b*c\$  | shift                      |
| \$aaAB*   | b*b*c\$   | Reduce $S \rightarrow aAB$ |
| \$aS*     | b*b*c\$   | shift                      |
| \$aS*b    | *b*c\$    | Reduce $S \rightarrow aAB$ |
| \$a*baAB  | *b*c\$    | shift                      |
| \$a*baAB* | b*c\$     | Reduce $S \rightarrow aAB$ |
| \$a*bsB*  | b*c\$     | shift                      |

\$a \* b S B \* b

\* C \$

Reduce  $S \rightarrow aAB$

\$ a \* b B \* b a AB

\* C \$

shift

\$ a \* b B \* b a AB \*

C \$

shift

\$ a \* b B \* S B \* c

\$

Accept

(4)

Stack

Input

Output

\$ S

abcc

shift

\$ Sa

\* b \* C \* C \$

Reduce  $S \rightarrow AB$

\$ a AB

\* b \* C \* C \$

shift

\$ a AB \*

b \* C \* C \$

Reduce  $aA \rightarrow aA$

\$ AB \*

b \* C \* C \$

shift

\$ AB \* b

\* C \* C \$

Reduce  $S \rightarrow AB$

\$ \* b S

\* C \* C \$

shift \*

\$ \* b S \*

C \* C \$

Reduce  $S \rightarrow AB$

\$ b AB

C \* C \$

Shift

\$ b ABC

\* C \$

$B \rightarrow cB$

\$ b AB

\* C \$

shift

\$ b AB \*

C \$

Reduce  $S \rightarrow AB$

\$ b S  
\$ b S C

C \$  
\$

Shift  
Accept

1b)

$$S' \rightarrow S$$

$$S \rightarrow aAB$$

$$A \rightarrow bA/c$$

$$B \rightarrow d$$

SLR Parsing

Ans.

 $I_0 :$ 

$$S' \rightarrow \cdot S$$

$$S \rightarrow \cdot aAB$$

$$A \rightarrow \cdot bA / \cdot c$$

$$B \rightarrow \cdot d$$

$$\text{Goto}(I_0, S) \equiv I_1 : S' \rightarrow S \cdot$$

$$S \rightarrow a \cdot AB \quad \text{closure}$$

$$A \rightarrow \cdot bA / \cdot c$$

$$B \rightarrow \cdot d$$

$$\text{Goto}(I_0, A) \equiv I_2$$

$$A \rightarrow b \cdot A$$

$$B \rightarrow \cdot d$$

$$\text{Goto}(I_0, B) \equiv I_3 : B \rightarrow d \cdot$$

$$\text{Goto}(I_1, A) \equiv I_4 : S \rightarrow aA \cdot B \quad A \rightarrow \cdot bA / \cdot c \quad B \rightarrow \cdot d$$

$$\text{Goto}(I_1, b) \equiv I_5 : A \rightarrow b \cdot A / \cdot c$$

$$B \rightarrow \cdot d$$

$\text{Goto } (I_1, d) \equiv I_6 : B \rightarrow d.$

$\text{Goto } (I_2, A) \equiv I_7 : A \rightarrow bA.$

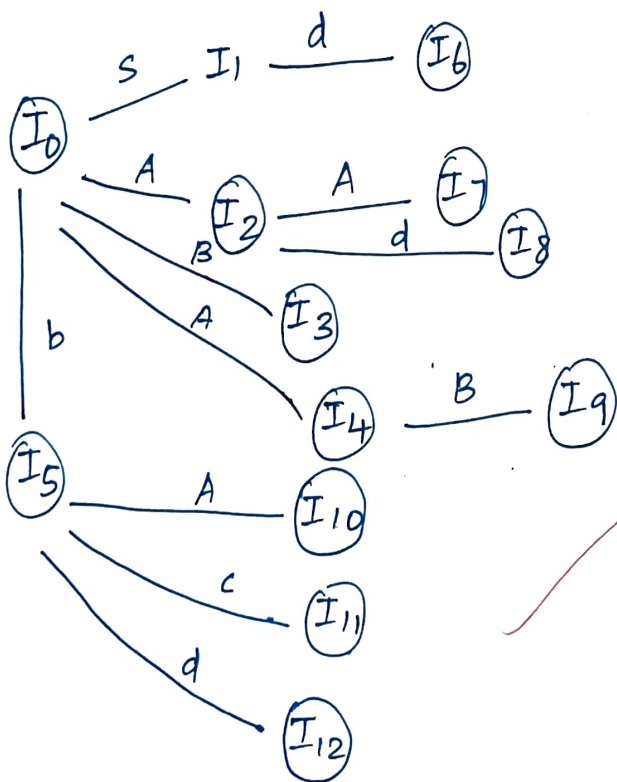
$\text{Goto } (I_2, d) \equiv I_8 : B \rightarrow d.$

$\text{Goto } (I_4, B) \equiv I_9 : S \rightarrow aAB.$

$\text{Goto } (I_5, A) \equiv I_{10} : A \rightarrow bA.$

$\text{Goto } (I_5, c) \equiv I_{11} : A \rightarrow c.$

$\text{Goto } (I_5, d) \equiv I_{12} : B \rightarrow d.$



18) Left Factoring:  $A \rightarrow aAB/aA/a$   
 $B \rightarrow bB/b$

$$A \rightarrow aAA'$$

$$A' \rightarrow AB/A/\epsilon$$

$$B \rightarrow bB'$$

$$B \rightarrow B/\epsilon$$

(17)  $S \rightarrow iEtss' / a, s' \rightarrow es/\epsilon, E \rightarrow b$

Left Recursion:

$$S \rightarrow aS'$$

$$S' \rightarrow iEtss'/\epsilon$$

$$S \rightarrow iEtss'/a$$

$$S' \rightarrow es/\epsilon$$

$$E \rightarrow b$$

$$\text{FIRST}(S) \rightarrow \{i, \epsilon, t\}$$

$$\text{FIRST}(S') \rightarrow \{e, \epsilon\}$$

$$\text{FIRST}(E) \rightarrow \{b\}$$

Non Terminal

id

S

S'

E

$$\text{FOLLOW}(S) \rightarrow \{\$, \epsilon\}$$

$$\text{FOLLOW}(S') \rightarrow \{\$, \epsilon\}$$

$$\text{FOLLOW}(E) \rightarrow \{\$, \epsilon\}$$

Terminal

\$

$$S \rightarrow iEtss'/a$$

$$S' \rightarrow es/\epsilon$$

$$E \rightarrow b$$

LL(1) Parsing Table :

$S \rightarrow ABC$      $A \rightarrow a/\epsilon$      $B \rightarrow \eta/\epsilon$

$S \rightarrow ABC$

$A \rightarrow a/\epsilon$

$B \rightarrow \eta/\epsilon$

$FIRST(S) \rightarrow \{a, \epsilon\}$

$FIRST(A) \rightarrow \{a, \epsilon\}$

$FIRST(B) \rightarrow \{\eta, \epsilon\}$

$FOLLOW(S) \rightarrow \{\$ \}$

$FOLLOW(A) \rightarrow \{\$ \}$      $FOLLOW(B) = \{\$ \}$

Terminal

Non Terminal

id

a

$\eta$

$\epsilon$

S

$S \rightarrow ABC$

$S \rightarrow ABC$

A

$A \rightarrow a$

$A \rightarrow \epsilon$

B

$B \rightarrow \eta$

$B \rightarrow \eta$

$B \rightarrow \epsilon$

20)  $S \rightarrow 1s\# / qABC$

$C \rightarrow b/\epsilon$

$A \rightarrow a/bbD$

$D \rightarrow c/\epsilon$

$B \rightarrow a/\epsilon$

Ans:

$S \rightarrow qABCS'$

$S' \rightarrow 1\#S' / \epsilon$

$A \rightarrow bbDA'$

$A' \rightarrow aA' / \epsilon$

$B \rightarrow a/\epsilon$

$C \rightarrow b/\epsilon$

$D \rightarrow c/\epsilon$

FIRST(S)  $\rightarrow \{q\}$

FIRST(S')  $\rightarrow \{1\}$

FIRST(A)  $\rightarrow \{a\}$

FIRST(B)  $\rightarrow \{a\}$

FIRST(C)  $\rightarrow \{b\}$

FIRST(D)  $\rightarrow \{c, \epsilon\}$

FOLLOW(S)  $\rightarrow \{\$ \}$

FOLLOW(S')  $\rightarrow \{\$, \#\}$

FOLLOW(A)  $\rightarrow \{\$, \#\}$

FOLLOW(B)  $\rightarrow \{\$, \#\}$

FOLLOW(C)  $\rightarrow \{\$, \#\}$

FOLLOW(D)  $\rightarrow \{\$, \#\}$

3

|              |  | Terminal |   |   |   |   |    |                           |
|--------------|--|----------|---|---|---|---|----|---------------------------|
| Non Terminal |  | d        | 1 | a | b | c | \$ | ε                         |
| S            |  |          |   |   |   |   |    |                           |
| S'           |  |          |   |   |   |   |    | $s \rightarrow qABCS'$    |
| A            |  |          |   |   |   |   |    | $s' \rightarrow \epsilon$ |
| B            |  |          |   |   |   |   |    | $A \rightarrow b b D A'$  |
| C            |  |          |   |   |   |   |    | $B \rightarrow \epsilon$  |
| D            |  |          |   |   |   |   |    | $C \rightarrow \epsilon$  |
|              |  |          |   |   |   |   |    | $D \rightarrow c$         |
|              |  |          |   |   |   |   |    | $D \rightarrow \epsilon$  |

## RUBRICS FOR CALCULATION

| Criteria                  | Weightage | Level 4 – Exemplary                                         | Level 3 – Proficient                    | Level 2 – Developing                         | Level 1 – Beginning                      |
|---------------------------|-----------|-------------------------------------------------------------|-----------------------------------------|----------------------------------------------|------------------------------------------|
| Understanding of Concepts | 25%       | Demonstrates thorough, accurate understanding of concepts   | Good understanding with minor gaps      | Basic understanding with some misconceptions | Poor understanding; major misconceptions |
| Explanation               | 25%       | Detailed, well-explained answers with relevant examples     | Clear explanation with adequate details | Limited explanation; lacks depth             | Poor or unclear explanation              |
| Application               | 20%       | Effectively applies concepts with strong analytical ability | Applies concepts with minor errors      | Limited application with weak analysis       | No application or incorrect analysis     |
| Organization              | 15%       | Well-structured answers with logical flow and coherence     | Organized with minor issues in flow     | Basic structure, lacks clarity               | Poor organization, incoherent answers    |

|          |     |                                                    |                                      |                                       |                                         |
|----------|-----|----------------------------------------------------|--------------------------------------|---------------------------------------|-----------------------------------------|
| Accuracy | 10% | Completely accurate and covers all required points | Mostly accurate with minor omissions | Partially correct with missing points | Incorrect answers with major gaps       |
| Clarity  | 5%  | Clear, neat, professional presentation             | Understandable with minor issues     | Limited clarity, readability issues   | Poorly written, difficult to understand |